Rewriting Expressions with Distributive Structure
Answer Key

Algebra 1   Expressions and Polynomial Operations

Quick Answers

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1. $12x + 30$
2. $36x + 135$
3. $12x - 54$
4. $18x + 24$
5. $x^2 + 11x + 24$
6. $14x + 8$
7. $5x\,(3x + 2)$
8. 25
9. $2x^2 + 15x + 18$
10. $24x + 27$, $24x + 27$

Curriculum

Level: Algebra 1   Expressions and Polynomial Operations
HSA-APR.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

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8. *If they got 37: distributed the 5 to the first term only and left the second term unchanged*

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1. Mural: height 6 ft, width $(2x + 5)$ ft. Write the area with no parentheses.

Step 1. Area of a rectangle is height times width, so the area is $6(2x + 5)$ square feet. The parentheses say "the whole width", so the 6 multiplies *both* pieces of that width.
Step 2. Distribute: $6 \cdot 2x = 12x$ and $6 \cdot 5 = 30$.
Step 3. The area is a sum of two rectangles, $12x$ and 30 square feet: $\boxed{12x + 30 \text{ ft}^2}$
Check by area model: a 6 by $2x$ strip beside a 6 by 5 strip has the same total area as one 6 by $(2x + 5)$ rectangle.

2. 9 members, each billed $(4x + 15)$ dollars. Total billing?

Step 1. Every member is billed the same expression, so the total is $9(4x + 15)$ dollars — a multiplier outside a sum, which is exactly the distributive setup.
Step 2. Distribute across both terms: $9 \cdot 4x = 36x$ and $9 \cdot 15 = 135$.
Step 3. Total billing in dollars: $\boxed{36x + 135}$
Reading the answer: $36x$ is the class charge for all nine members and 135 is the fixed part of the bill.

3. **6 boards, each $(2x-9)$ inches long. Total length?**
Step 1. The length repeated six times is $6(2x-9)$ inches. The minus sign belongs to the 9, so the second product will be subtracted.
Step 2. $6 \cdot 2x = 12x$, and $6 \cdot (-9) = -54$.
Step 3. Total length: $\boxed{12x - 54 \text{ in}}$
Common wrong answer: $12x-9$ — the 6 was multiplied into the first term only. Distribution reaches every term inside the parentheses.

4. **Panel area $(18x+24)$ cm^2, height 6 cm. Write $6(\quad)$.**
Step 1. Factoring is distribution run backwards: we need the number that multiplies into both $18x$ and 24. The height 6 is given, so test 6 as the common factor.
Step 2. $18x \div 6 = 3x$ and $24 \div 6 = 4$, so the width is $3x+4$ centimetres.
Step 3. Factored form: $\boxed{6(3x+4)}$
Check: distribute back — $6 \cdot 3x + 6 \cdot 4 = 18x + 24$, the area we started with.

5. **Patio $(x+3)$ ft by $(x+8)$ ft. Write the area with no parentheses.**
Step 1. Both factors are sums now, so the distributive property is applied twice: every term of the first factor multiplies every term of the second.
Step 2. $x \cdot x = x^2, \quad x \cdot 8 = 8x, \quad 3 \cdot x = 3x, \quad 3 \cdot 8 = 24.$
Step 3. Combine the two like terms $8x + 3x = 11x$: $\boxed{x^2 + 11x + 24 \text{ ft}^2}$
Area model: the four products are the four rectangles of the patio — the original x by x square, two strips, and the 3 by 8 corner.

6. **Is $4(3x+2)+2x$ the same as $14x+8$?**
Step 1. A claimed equivalence is settled by rewriting one side into simplest form, so expand the left side first.
Step 2. $4(3x+2) = 12x+8$. Now add the loose term: $12x + 8 + 2x$.
Step 3. Combine like terms: $12x + 2x = 14x$, and the constant 8 stays. $\boxed{14x+8}$
Maya is correct: the two expressions are equivalent, because the left side simplifies to exactly $14x+8$.

7. **Area $(15x^2+10x)$ square units, width $5x$ units. Factor.**
Step 1. The width is given as $5x$, so $5x$ should come out of both terms — take the greatest common factor of the numbers (5) together with the highest power of x in every term (x).
Step 2. $15x^2 \div 5x = 3x$ and $10x \div 5x = 2$, so the length is $3x+2$ units.
Step 3. Factored completely: $\boxed{5x(3x+2)}$
Common wrong answer: $5(3x^2+2x)$ — the numerical factor came out but the shared x did not, so the factoring is not complete.

8. **Devin wrote $5(2x-3) = 10x-3$. Evaluate the original at $x=4$.**
Step 1. Two expressions are equivalent only if they agree at *every* value of x, so one value that disagrees settles it.
Step 2. Substitute into the original: $5(2 \cdot 4 - 3) = 5(8-3) = 5 \cdot 5$.
Step 3. The original expression is worth $\boxed{25}$
Devin's form gives 37 at $x=4$, so the forms are not equivalent. The mistake: the 5 was distributed to $2x$ only, leaving the -3 untouched. Distributing to both terms gives $10x-15$, and $10(4)-15 = 25$ as it should.

9. **Rug** $(2x+3)$ **ft by** $(x+6)$ **ft. Write the area with no parentheses.**
Step 1. Two binomial factors again, but the first one now has a leading coefficient of 2 — every term still multiplies every term.
Step 2. $2x \cdot x = 2x^2, \quad 2x \cdot 6 = 12x, \quad 3 \cdot x = 3x, \quad 3 \cdot 6 = 18.$
Step 3. Combine $12x + 3x = 15x$: $\boxed{2x^2 + 15x + 18 \text{ ft}^2}$
Sanity check: the leading term $2x^2$ has to come from the two leading terms, and $2x \cdot x$ is the only way to get it.

10. 6 **kits at** $(2x+5)$ **dollars and** 3 **kits at** $(4x-1)$ **dollars. (a) Total as** $ax+b$**. (b) Factor it.**
Step 1 (a). Two separate groups, so distribute each multiplier over its own parentheses before anything is combined.
Step 2 (a). $6(2x+5) = 12x + 30$ and $3(4x-1) = 12x - 3$.
Step 3 (a). Add and collect like terms: $12x + 12x = 24x$ and $30 - 3 = 27$. $\boxed{24x + 27}$
Step 4 (b). Now run distribution backwards on $24x + 27$. The numbers 24 and 27 share a factor of 3, and there is no x in the constant term, so the greatest common factor is just 3.
Step 5 (b). $24x \div 3 = 8x$ and $27 \div 3 = 9$: $\boxed{3(8x+9)}$
Why 3 *and not* 9*:* 9 divides 27 but not 24, so it is not common to both terms.